

Individual Contribution Statement

Manuscript: CALiBRE-AD+: A Nested, Leakage-Audited, Calibrated, Cost-Sensitive Ensemble for Alzheimer’s Disease Classification **Author:** MD Rahat Rihan (sole author), Department of Computer Science, American International University-Bangladesh

CRedit-style contribution breakdown

Because this is a single-author manuscript, a multi-person split is not applicable — but the rubric asks for contributions to be named and cross-checkable against commit history, so we break the work down by *role* rather than by *person*, and separately disclose which parts had AI assistance (see 07_integrity_disclosure/AI_Tool_Usage_Disclosure.md for the full breakdown).

Role (CRedit taxonomy)	Contributor	Notes
Conceptualization	MD Rahat Rihan	Original hypothesis: headline AD benchmark accuracy is driven by leakage from clinician-administered assessment features.
Data curation	MD Rahat Rihan	Selected and hashed (MD5) the dataset copy; verified provenance against the Kaggle source.
Methodology	MD Rahat Rihan	Designed the nested 5×3 cross-fitting protocol, the Full/Screening track split, the seven-learner stacking ensemble, the adaptive calibration scheme, and the cost-sensitive threshold search.
Software	MD Rahat Rihan	Wrote and ran <code>alzheimerupdate_version_1.ipynb</code> , including all model training, evaluation, bootstrap CI, McNemar tests, and SHAP analysis.
Validation	MD Rahat Rihan	Identified and removed fabricated content from an earlier AI-drafted version of the paper (a phantom ablation table, an unverified learning-curve claim, an unsupported McNemar comparison) prior to this revision cycle.
Formal analysis	MD Rahat Rihan	Interpreted the Full-vs-Screening collapse, the calibration results, and the ablation study.
Writing – original draft	MD Rahat Rihan	Authored the original manuscript content and all reported claims.
Writing – review & editing (this revision cycle)	MD Rahat Rihan + Claude (AI assistance)	See 07_integrity_disclosure/AI_Tool_Usage_Disclosure for the exact division.

Role (CRediT taxonomy)	Contributor	Notes
Visualization	MD Rahat Rihan (Figs. 3–10, notebook-generated) + Claude (Fig. 2, architecture diagram, AI-generated script)	
Supervision / Project administration	MD Rahat Rihan	Sole author; no co-authors or supervisors listed on the manuscript.

Cross-checking against commit history

The manuscript’s code and notebook are hosted at <https://github.com/rahatRiSD/CSAA-MSCS->. At the time this package was assembled, that repository was confirmed (via direct inspection) to contain: `README.md`, `alzheimer-disease classification.ipynb`, `alzheimer-update (1).ipynb`, and `alzheimers_disease_data_loaded.csv`, across **3 commits** on the main branch, all attributable to a single GitHub account (the author’s).

Limitation of this statement: this AI assistant does not currently have authenticated access to the repository’s commit API from within this session, so the exact commit SHAs, timestamps, and per-commit messages could not be pulled and verified programmatically here. To make this section fully cross-checkable as the rubric requires, the author should run the following locally and paste the output below before submission:

```
git clone https://github.com/rahatRiSD/CSAA-MSCS-.git
cd CSAA-MSCS-
git log --pretty=format:"%h | %an | %ad | %s" --date=short
```

→ [Paste git log output here before final submission] →

Since this is single-author work, all commits should show the same author identity as the paper’s byline (MD Rahat Rihan); if any commit shows a different author (e.g., a collaborator or a different account), that should be reconciled and named explicitly here rather than left unexplained, since an unexplained mismatch is exactly what this rubric item is designed to catch.

Statement of accuracy

The author affirms that the contribution breakdown above accurately reflects who did what, and that the AI-assistance boundary matches the disclosure in `07_integrity_disclosure/AI_Tool_Usage_Disclosure.md`.